Nader Khalil

Full-Stack AI Engineer

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PROFESSIONAL SUMMARY

Full-Stack AI software engineer with a B.Sc. in Software Engineering and Unix training from 42 Abu Dhabi, building web applications and production AI systems with exceptional velocity. Combines full-stack architecture (Next.js 16, React 19, PHP 8.4, PostgreSQL) with applied AI integration (RAG, tool calling, prompt caching) and modern AI-accelerated workflows (Cursor, Claude, Copilot). Backed by rigorous engineering discipline: database row locks, strict type safety, and 766+ automated tests in CI. Bilingual in Arabic (Native) and English (Professional), based in Abu Dhabi.

TECHNICAL SKILLS

Full-Stack AI Delivery: Applied GenAI integration (RAG, tool calling, prompt caching), AI-assisted engineering (Cursor, Claude Code), rapid prototyping to production

Backend & APIs: RESTful APIs, layered architecture, PHP 8.4, C# MinimalAPI, Zod schema validation, Auth.js, RBAC, input sanitization

Testing & CI/CD: 766+ automated tests in CI, Vitest, Playwright (e2e), PHPUnit, concurrent race testing, GitHub Actions, Docker

Frontend Engineering: React 19, Next.js 16 (App Router), TypeScript, Tailwind CSS, Vite, TanStack Query, accessible UI design (axe-core)

Data & Concurrency: PostgreSQL 16 + pgvector, MariaDB, Prisma ORM, row-level locks (`SELECT FOR UPDATE`), transaction isolation, CSP/CSRF

Languages & Core: TypeScript, JavaScript, Python, PHP, C, C++, C#, SQL, Bash, Linux system internals

TECHNICAL PROJECTS

AI Booking Platform, *Production Multi-Tenant Web Platform* 2026
Live build: booking.naderkhalil.dev · Case study: naderkhalil.dev/case-study.html

- Architected and shipped a full-stack booking system end-to-end: live calendars, advisor availability rules, admin knowledge bases, and multi-tenant access control (Next.js 16, TypeScript, PostgreSQL + pgvector, Prisma, Auth.js).
- Eliminated race conditions with database row-level locking on PostgreSQL; backed by 766 automated tests (357 unit, 320 integration, 89 e2e Playwright) and 93% cached input inference efficiency.
- Integrated an in-context RAG assistant proposing calendar mutations as interactive confirmation cards requiring explicit human review.

Student-Faculty Services v2, *High-Performance Service Platform* 2026
Live build: student-faculty-services-v2.infinityfree.io · Case study: naderkhalil.dev/case-study-v2.html

- Architected a framework-free PHP 8.4 REST API (46 endpoints) and React 19 frontend with 10-second cancellation undo, reminders, waitlists, and 0 double bookings across 8-student concurrent race tests; verified by 500+ tests with 100/100 accessibility.

42 Intra Logtime Calculator, *Published Browser Extension* 2026
Chrome Web Store: chromewebstore.google.com

- Lightweight Chrome extension (19 KB, 60+ users, 5.0★) delivering real-time calendar logtime calculations client-side with zero data collection across both legacy and V3 42 Intra layouts.

PROFESSIONAL EXPERIENCE

AI Trainer & Mentor, *Department of Education and Knowledge (ADEK)* 07/2026 – 08/2026

- Delivered technical instruction on agentic workflows, multi-agent orchestration, and autonomous tool calling in n8n for the accredited AgenticVerse program; mentored teams building AI prototypes for ADNOC, FANR, and Khalifa Fund.
- Guided participants through API schema validation, structured output parsing, environment configuration, and debugging in Python and TypeScript.

QA & Software Tester, *42 Abu Dhabi* 01/2026

- Conducted functional and UI testing on internal academic platforms using black-box scenarios; authored reproduction reports with logs and steps to help developers isolate defects quickly.
- Tested features across browsers and viewports; re-tested defect fixes across user flows to ensure zero regressions before release.

IT Support & Technical Staff, *Ticketmaster* 12/2025

- Provisioned and deployed entry ticketing hardware and access scanners for a high-volume international event; maintained 100% operational uptime through real-time hardware health checks and rapid on-site incident resolution.

Software Engineering Intern, *Timex IT Consulting* 01/2024 – 04/2024

- Developed C# MinimalAPI CRUD endpoints with request validation and standardized HTTP response handling; verified functional defect fixes on client accounting software before production release.

EDUCATION & CERTIFICATIONS

B.Sc. in Software Engineering, *Al Ain University*
Computer Science Diploma, *42 Abu Dhabi*
MOEI × 42 Abu Dhabi AgentEra Hackathon, *Honoree*

CyberQ CTF, Top 10 Finalist, *UAE Cybersecurity Council*
Hack The Box University CTF, *Top 12% (119/1,014 teams)*
RecognAIze Hackathon, *AI on Watch with Datamind*

Languages: Arabic (Native), English (Professional Proficiency)